LAB 1

# GEOSPATIAL REPRESENTATION LEARNING

## Visualizing high-dimensional representation spaces

PRESS → OR SPACE.

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# Recap: Lecture 1

## PCA

A linear projection that captures variance.

Rotate to PCA projection | Reset | ☑ Show projections

## t-SNE

A nonlinear embedding that preserves local neighborhoods.

Run t-SNE projection | Reset | ☑ Show trajectories

Loading slide...

# Expert Group Activity

Each group becomes an expert in **one method**, then teaches a group studying the other topic.

## PCA

**Groups 1, 3, 5, 7, 9**

Principal Component Analysis

## t-SNE

**Groups 2, 4, 6, 8, 10**

t-distributed Stochastic Neighbor Embedding

**Teaching pairs:** 1 & 2, 3 & 4, 5 & 6, 7 & 8, 9 & 10

# Lab schedule

| Time | Activity | Result |
|---|---|---|
| **60 min** | Self-teaching in expert groups | Understand your method and prepare your group’s slides |
| **45 min** | Teach the group studying the other method | Both methods, with time for questions and comparison |
| **15 min** | Class-wise synthesis | Discuss exam questions selected by the teacher |

**Total: 120 minutes**

# Expert group preparation

During the **60-minute preparation phase**, use the notebook and your group's slide deck.

- Explain your method's core idea and how it constructs a low-dimensional representation
- Explore the GloVe embeddings and include plots that support your explanation
- Discuss what the plots reveal, the method's limitations and when to use it
- Develop **two exam questions with proposed answers** and add them to your deck

**The teacher will select questions from your decks for the final class discussion.**

Share the teaching across your group. You have 45 minutes together with your partner group.

# Word embeddings with GloVe

Open the Google Colab notebook to **explore PCA and t-SNE in practice**.

GloVe represents words as vectors. Here, both methods project **300-dimensional word embeddings into 2D**.

## Principal Component Analysis (PCA)

## t-distr. Stochastic Neighborhood Embedding

# Group slide decks

Open your assigned deck and add your group's explanations, plots and exam questions directly in Google Slides.

Shared Google Drive folder

| Group | Method | Slides | Group | Method | Slides |
|---|---|---|---|---|---|
| 1 | PCA | Edit slides | 2 | t-SNE | Edit slides |
| 3 | PCA | Edit slides | 4 | t-SNE | Edit slides |
| 5 | PCA | Edit slides | 6 | t-SNE | Edit slides |
| 7 | PCA | Edit slides | 8 | t-SNE | Edit slides |
| 9 | PCA | Edit slides | 10 | t-SNE | Edit slides |

# Phase 1: Self-teaching

**13:00–14:00 · 60 minutes**

Work on your group's slide deck. Use the notebook to explore your method and prepare your explanation and plots.

**Add two exam questions with proposed answers.** The teacher will select questions for the final discussion.

| Group | Method | Slides | Group | Method | Slides |
|---|---|---|---|---|---|
| 1 | PCA | Edit slides | 2 | t-SNE | Edit slides |
| 3 | PCA | Edit slides | 4 | t-SNE | Edit slides |
| 5 | PCA | Edit slides | 6 | t-SNE | Edit slides |
| 7 | PCA | Edit slides | 8 | t-SNE | Edit slides |
| 9 | PCA | Edit slides | 10 | t-SNE | Edit slides |

# Phase 2: Teach the other group

**14:00–14:45 · 45 minutes**

## Teaching pairs

| PCA group | t-SNE group |
|---|---|
| 1 | 2 |
| 3 | 4 |
| 5 | 6 |
| 7 | 8 |
| 9 | 10 |

## Both groups teach

- Use your slides to explain your method and demonstrate your plots
- Give each group time to teach and answer questions
- Compare what PCA and t-SNE reveal and discuss their limitations

# Phase 3: Class-wise synthesis

**14:45–15:00 · 15 minutes**

The teacher asks selected exam questions from the group decks.

- Discuss the questions together as a class
- Explain the reasoning behind your answers
- Connect the answers to the strengths and limitations of PCA and t-SNE

**Use this discussion to clarify anything that remains unclear.**

# Revisit the learning outcomes

Light gray fill: your group's before responses today · Click to respond again.